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Palaeomagnetic evidence for post-thrusting tectonic rotation in the Southeast Pyrenees, Spain

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Abstract

The structural framework of the Southeast Pyrenees led to two conflicting interpretations—thrust tectonics vs. wrench tectonics—to explain the geometry of this mountain range. In the present study palaeomagnetic data are presented in an attempt to resolve this conflict. The data reveal different magnetisation directions that indicate tectonic rotations about vertical axes. By means of a regionally homogeneous pattern of rotation, three tectonic units could be distinguished in the Southeast Pyrenees. The Internal Unit in the north reveals no rotation since the Permian. The External Unit to the south shows anticlockwise rotation of 25°, younger than the Early Oligocene. The Pedraforca Unit, placed on the External Unit, shows 57° clockwise rotation which can be assigned to the Neogene. The anticlockwise rotation of the External Unit can be explained by differential compression during the last phase of Pyrenean thrusting, whereas the clockwise rotation of the Pedraforca Unit can be interpreted by post-thrusting tectonics. The rotation pattern of the Southeast Pyrenees provides evidence for both Cretaceous to Paleogene N–S compression and Neogene right-lateral wrench tectonics.

1. Introduction

The Pyrenees are part of the Alpine orogen and formed mainly during the Late Cretaceous to Neogene. They can be subdivided into the North and South Pyrenees, two marginal thrust belts separated by the Axial Zone (Muñoz, 1992). The tectonic framework of the Pyrenees is dominated by Alpine thrusts, but Neogene structures also contribute to the present-day shape of the orogen (e.g., Pous et al., 1986). In the Axial Zone, pull-apart basins, which are a result of right-lateral slip along the Catalan fault (Cabrera et al., 1988), document such Neogene deformation (Fig. 1a). Thus far, little attention has been paid to the influence of Neogene tectonics on the geometry of the marginal thrust belts of the Pyrenees.

In the southern thrust belt, the Segre–Transform Zone (STZ) has been considered as a continuation of the Catalan fault (Fig. 1) (Souquet, 1977). The STZ separates the South Pyrenean thrust belt tectonically into a central and an east-

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ern part (e.g., Souquet, 1977). The central part consists of three main thrust sheets forming the South Central Unit (Seguret, 1972). The eastern part is built up by the Pedraforca and Cadí units (Muñoz et al., 1986). The Pedraforca Unit can be described as the stratigraphic and tectonic equivalent to the South Central Unit (e.g., Ullastre et al., 1987; Vergés et al., 1992). By contrast, the Cadí Unit has no tectonic equivalent in the central part of the Pyrenees (Fig. 1).

Two principal models have been put forward to explain the restricted occurrence of the Cadí

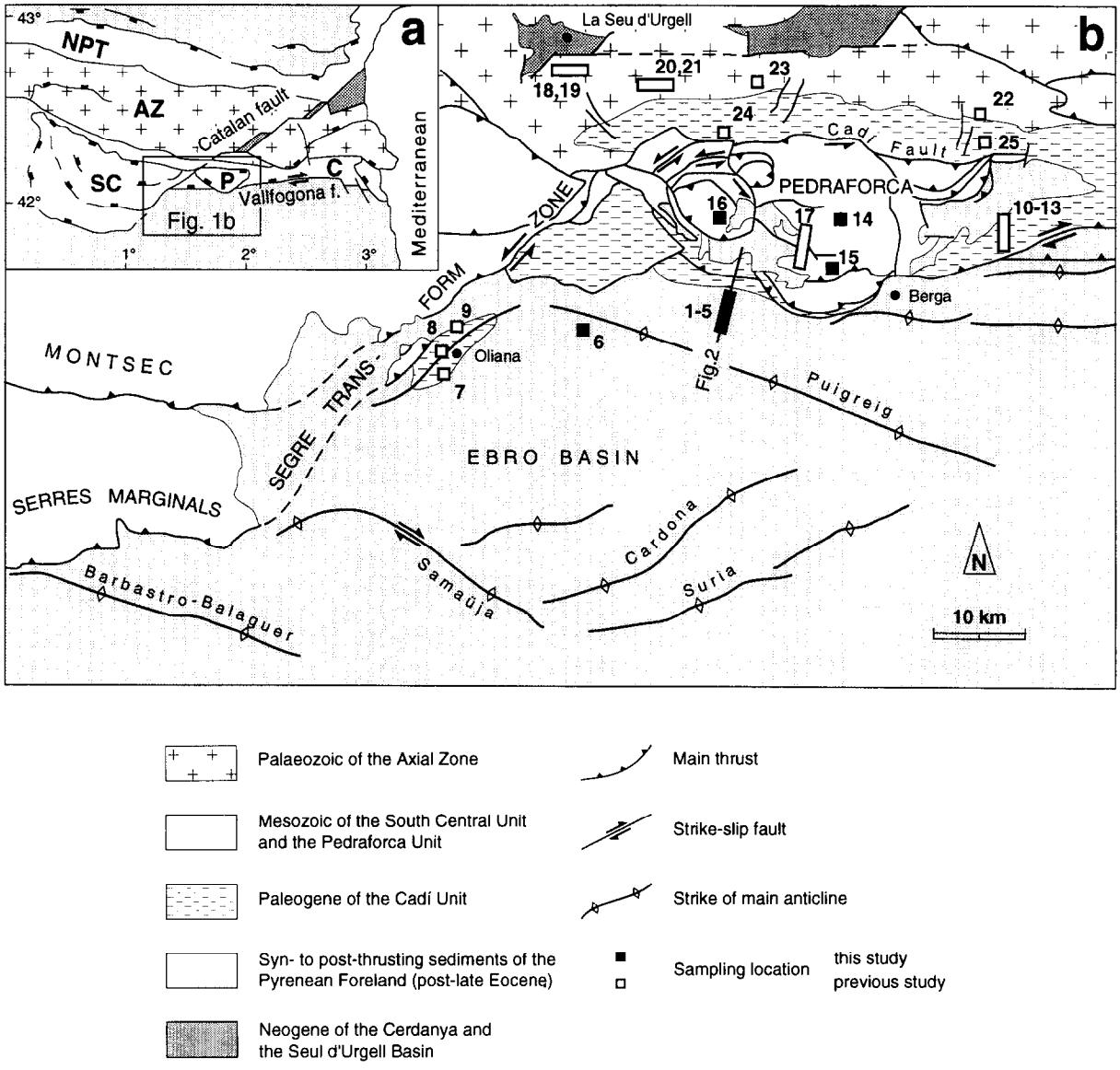


Fig. 1. (a) Overview of the East Pyrenees with the major tectonic units, the Axial Zone (*AZ*), the North Pyrenean thrust belt (*NPT*), and the South Pyrenean thrust belt (*SC* = South Central Unit, *P* = Pedraforca Unit, *C* = Cadí Unit). (b) Structural setting of the central and the eastern part of the South Pyrenees separated by the Segre–Transform Zone, and their foreland.

Unit. One model invokes the Cadí Unit as a thrust sheet emplaced southwards during Alpine tectonics (e.g., Muñoz et al., 1986; Vergés and Muñoz, 1990; Vergés et al., 1992). The other model, however, challenged thin-skinned thrusting of the Cadí Unit, and explains the contact between the two units of the Southeast Pyrenees by means of Neogene wrench tectonics (Durand-Delga et al., 1989), which formed the steep Cadí and Vallfogona faults (Fig. 1). If the Cadí Unit is autochthonous then the amount of shortening in the East Pyrenees has to be less than that proposed in the thrust model.

Palaeomagnetic data from the Pyrenean foreland (Dinarès et al., 1992) and the Pedraforca Unit (Keller and Gehring, 1992) have shown contrasting rotation of single anticlines, which can be interpreted by tectonic rotation about vertical axes. None of the above models (i.e., thrust tectonics vs. wrench tectonics) has considered these rotations. The aim of this study is to present regionally distributed palaeomagnetic data, in order to develop a deformation scenario for the Southeast Pyrenees that includes tectonic rotation.

2. Structural setting and timing of deformation in the Southeast Pyrenees

The Pedraforca Unit can be subdivided into an upper and a lower thrust sheet based on its Cretaceous stratigraphy (Vergés and Martínez, 1988). The upper Pedraforca thrust sheet is dominated by Early Cretaceous limestone. By contrast, the lower thrust sheet is mainly built up by the Bona Formation, a thick sequence of Late Cretaceous shelf limestone (Mey et al., 1968). During the latest Cretaceous and the earliest Tertiary the upper thrust sheet was emplaced onto the Bona Formation (Ullastre et al., 1987). During the Early Eocene, the two thrust sheets of the Pedraforca Unit were emplaced onto the Paleogene Pyrenean foreland, whereas the upper thrust sheet was passively transported, riding on the lower one (Vergés and Martínez, 1988). At Aigua de Valls (Fig. 2) the end of this tectonic phase can be dated as Lutetian by flysch sediments overlying an erosional discordance (horizon A in Fig. 2; Solé-Sugrañes, 1972).

Subsequent N–S compression led to deformation of the Paleogene foreland (Puigdefàbregas et

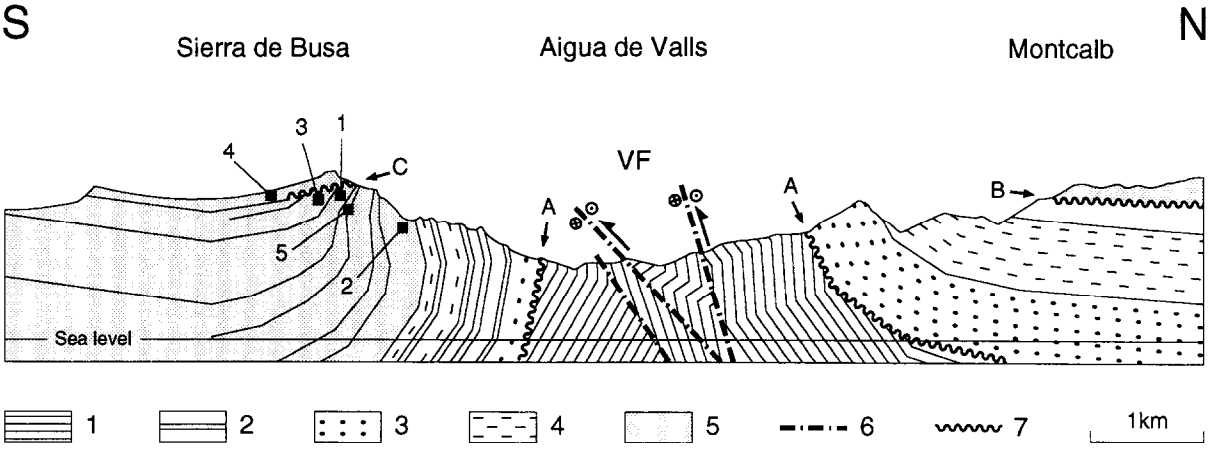


Fig. 2. Schematic cross-section, with sampling locations 1–5 in the Sierra de Busa (after Solé-Sugrañes, 1973; Riba, 1976; Masriera and Ullastre, 1985). Cadí Unit: *1* = marls (early Lutetian); *2* = sandstones; *3* = conglomerates and *4* = marls (Flysch sediments of the late Lutetian–Bartonian). Syn- to post-thrusting sediments: *5* = Berga Formation (Bartonian–Stampian). *6* = tectonic contacts. *7* = erosional discordances: *A* = Lutetian; *B* = post-late Bartonian; *C* = Early Oligocene. *VF* = Vallfogona fault. For location of the cross section see Fig. 1b.

al., 1986) that is well documented by tilted flysch sediments and syn-tectonic conglomerates of the Berga Formation in the Sierra de Busa (Fig. 2). Muñoz et al. (1986) linked the deformation of the Paleogene foreland to the thrusting of the sediments in the Sierra del Cadí. Based on this structural correlation, they identified the Cadí Unit as a thrust sheet (Fig. 1). In the Sierra de Busa (Fig. 2), the end of deformation related to the emplacement of the Cadí Unit is documented by an erosional discordance in the Berga Formation dated as Early Oligocene (horizon C in Fig. 2) (Riba, 1976). Shortening during middle and Late Oligocene led to folds and thrusts in the Ebro Basin (Vergés et al., 1992). The Sanaüja and Suria anticlines mark the southern front of this deformation (Fig. 1b), and are the youngest structures related to Pyrenean thrust tectonics.

At Montcalb the conglomerates of the Berga Formation cover an erosional horizon on the Cadí Unit (horizon B in Fig. 2). Masriera and Ullastre (1985) dated this erosion as younger than late Bartonian. To the north the erosional horizon can be followed onto the Pedraforca Unit (Fig. 1b). The conglomerates of the Berga Formation seal the tectonic contact between the two units. In the Pedraforca Unit the conglomerates are often folded (Keller and Gehring, 1992) or overthrust by Mesozoic sediments (Guérin-Desjardins and Latreille, 1961). In the northern part of the Pedraforca Unit, strike-slip faults dissect the post-Bartonian conglomerates as well as the thrust structures (Fig. 1b; Solé-Sugrañes, 1973; Ullastre et al., 1987; Durand-Delga et al., 1989). This deformation is indicative of post-thrusting tectonics in the Southeast Pyrenees.

3. Samples and palaeomagnetic methods

Oriented samples were collected in sandstone and siltstone of the Late Eocene to middle Oligocene Berga Formation (Fig. 1b; locations 1–6) and in limestone of the Late Cretaceous Bona Formation (Fig. 1b; locations 14–16). The palaeomagnetic samples were drilled in the field, and 2.5 cm diameter cores were oriented using a magnetic compass and an inclinometer. The natural remanent magnetisation (NRM) was measured with a 2G three-axes cryogenic magnetometer. Progressive thermal demagnetisation was performed using a Schonstedt oven. The charac-

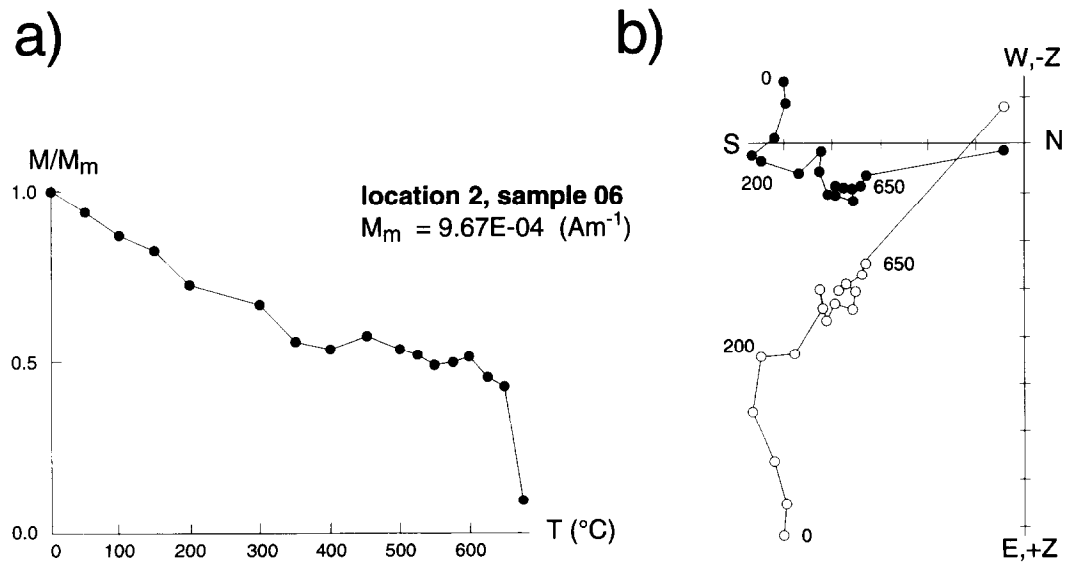


Fig. 3. (a) Decrease in intensity and (b) vector plots of the natural remanence (before tectonic correction) during thermal demagnetisation of a sandstone from the Berga Formation. Full and open symbols present projections of the magnetisation vector on the horizontal and N–S vertical plane.

teristic NRM components were evaluated by least-squares fitting of linear segments to orthogonal vector plots during progressive demagnetisation (Zijderveld, 1967). Isothermal remanent magnetisation (IRM) in fields up to 1 T, followed by thermal demagnetisation of a two-component IRM, was used to identify magnetic minerals in the samples by means of coercivity and unblocking temperature spectrum analysis (Lowrie and Heller, 1982; Lowrie, 1990).

4. Palaeomagnetic and rock magnetic results

4.1. The Berga Formation

The intensity of the NRM in the sandstone and siltstone varies between 0.30 and 2.00 mA m^{-1}. During thermal demagnetisation, the NRM is observed to consist of at least two directional components (Fig. 3). A low-temperature component is demagnetised below 200°C. The second component shows a minor intensity decrease up to 650°C and a distinct unblocking between 650 and 675°C. Isothermal remanent magnetisation of unheated samples is dominated by high-coercivity minerals, which are far from saturation in the maximum fields applied (Fig. 4a). The thermal demagnetisation of the hard IRM component, acquired in a 1 T field, shows a distinct unblocking step around 100°C, followed by a steady decrease of the IRM intensity, and a second unblocking step between 700 and 725°C (Fig. 4b). The thermal demagnetisation of the soft IRM component acquired in a 0.12 T field is similar to the high-coercivity fraction, but without the component below 100°C, which may be due to goethite, α-FeOOH (Lowrie and Heller, 1982). The direction of this low-temperature component, however, is preserved up to 200°C (Fig. 3b), well above the Néel point of goethite. Thus, it is very likely that fine-grained hematite (α-Fe_2O_3) also contributes to the low-temperature component (Purucker et al., 1980). The high-temperature NRM component which is only completely demagnetised above 675°C, can be attributed to specularite (α-Fe_2O_3) (Purucker et al., 1980). Goethite and fine-grained hematite for the low-temperature component, and specularite for the high-temperature component are therefore considered as the carriers of the remanence of the rock samples.

In the Sierra de Busa (Figs. 1b and 2; locations 1–5) and the Puigreig anticlines (Fig. 1b; location

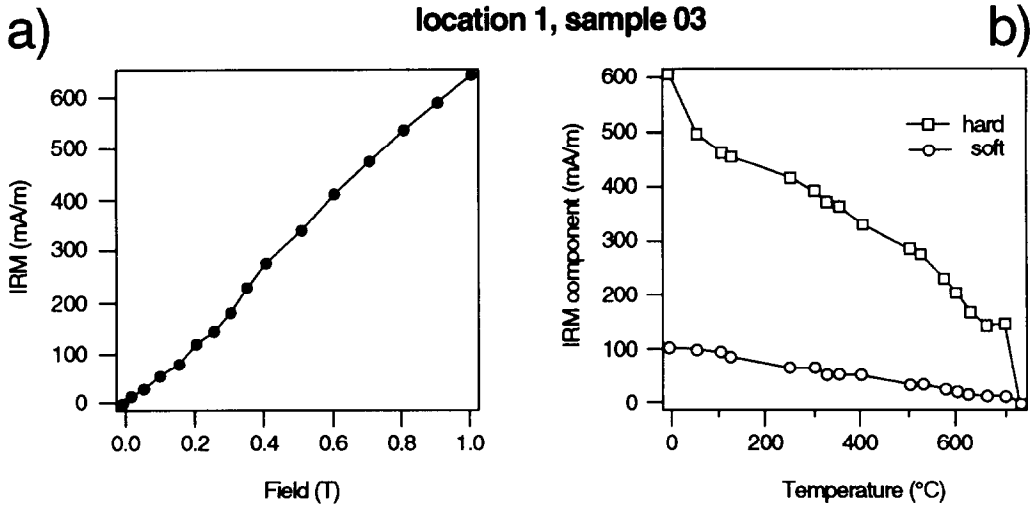


Fig. 4. (a) IRM acquisition and (b) thermal demagnetisation of a two-component IRM of a sandstone from the Berga Formation. The two-component IRM is acquired by magnetising the sample in an 1 T field (hard) along its z-axis, and in a 0.12 T field (soft) along its x-axis (Lowrie, 1990).

6) the folded Berga Formation permits a comparison between the mean remanence directions before and after tectonic correction (Table 1; Fig. 5). The location mean of the low-temperature component before tectonic correction shows a declination of 331.7° and an inclination of 59.4° with a precision parameter k (Fisher, 1953) of 21.3 (Table 1; Fig. 5a). Application of tectonic correction increases the scatter of the mean remanence directions (Fig. 5b), and is indicated by a lower k-value of the location mean (Table 1; $k = 5.5$). The comparison of the statistical f-parameters (McFadden and Jones, 1981) before and after tectonic correction indicates a significant negative fold test, and that points to post-folding origin of the goethite/hematite magnetisation.

The mean directions of the high-temperature component are scattered before tectonic correction (Table 1; Fig. 5c; $k = 4.6$). After tectonic correction (Fig. 5d) the precision parameter for the location mean ($k = 11.7$) increases by more than a factor of two. The McFadden and Jones (1981) fold test, however, does not show a significant difference between both directions. A comparison of the inclinations of the locations (Fig. 5c and d) shows a tighter grouping after tectonic correction, and that argues in favour of a remanence acquisition during sedimentation. The location mean of the high-temperature component

Table 1
Sampling locations and mean palaeomagnetic directions

Location	Demag. range (°C)	n/r	D_{BTC}	I_{BTC}	α_{95}	k	f_{BTC}	D_{ATC}	I_{ATC}	α_{95}	k	f_{ATC}
Berga Formation												
1	25–300	10/0	330.7	65.0	5.0	93.17		251.6	60.9	5.1	90.3	
	300–650	0/11	180.2	−37.7	12.9	13.64		152.0	−61.5	13.7	12.1	
2	25–300	11/0	351.5	54.3	7.6	47.0		203.6	27.2	7.7	46.2	
	300–650	0/10	162.5	59.4	20.5	6.5		171.7	−34.6	10.4	22.7	
3	25–200	8/0	330.2	51.8	15.8	15.6		267.1	67.3	22.4	8.2	
	300–650	9/0	336.6	31.0	13.5	16.5		310.7	52.4	11.0	16.3	
4	25–300	10/0	329.0	63.6	12.4	16.2		316.7	71.0	20.0	6.8	
	400–650	6/3	346.5	47.7	9.3	31.3		341.2	59.8	8.9	34.2	
5	25–300	10/1	316.7	62.4	8.8	31.4		225.0	51.5	8.9	30.2	
	300–650	7/0	357.7	5.2	14.7	17.9		349.1	43.5	16.8	13.9	
6	25–200	8/0	329.2	52.7	15.7	13.5		337.1	46.9	14.7	15.1	
	300–600	0/5	168.3	−46.1	12.2	40.3		173.7	−38.4	15.7	42.6	
Mean 1–6	25–300	57/1	331.7	59.4	4.3	21.3	2.26 (1)	245.8	62.9	9.1	5.5	100.98 (1)
	300–650	22/29	349.7	25.0	10.5	4.6	21.60 (2)	340.4	50.5	6.1	11.7	4.72 (2)
Bona Formation												
14	25–100	5/0	359.1	66.9	11.6	44.6		14.4	34.7	14.7	28.2	
	200–650	9/1	24.5	50.5	6.9	44.4		24.6	22.3	6.0	59.3	
15	25–100	4/0	6.9	65.4	14.9	39.0		0.7	24.6	13.8	45.1	
	200–600	0/8	230.8	−40.1	12.4	21.0		215.4	−10.3	12.5	20.7	
16	25–125	4/2	340.7	14.7	30.7	5.7		345.7	6.5	29.9	6.0	
	200–600	7/3	27.2	35.4	20.0	6.8		21.3	1.2	20.1	6.7	
Mean 14–16	200–650	16/12	35.3	42.4	7.7	13.5	0.94 (3)	25.9	11.8	8.0	12.6	2.11 (3)

Locations 1–6 are from the Berga Formation in the Pyrenean foreland, and locations 14–16 from the Bona Formation in the Pedraforca Unit. (n) is the number of samples with normal and (r) with reversed polarity. Declination (D) and inclination (I) with BTC and ATC denote location mean directions, before and after tectonic correction, respectively.
Fold tests are significant at the 99% confidence level when $f < 2.51$ (1), 2.55 (2), or 3.74 (3).

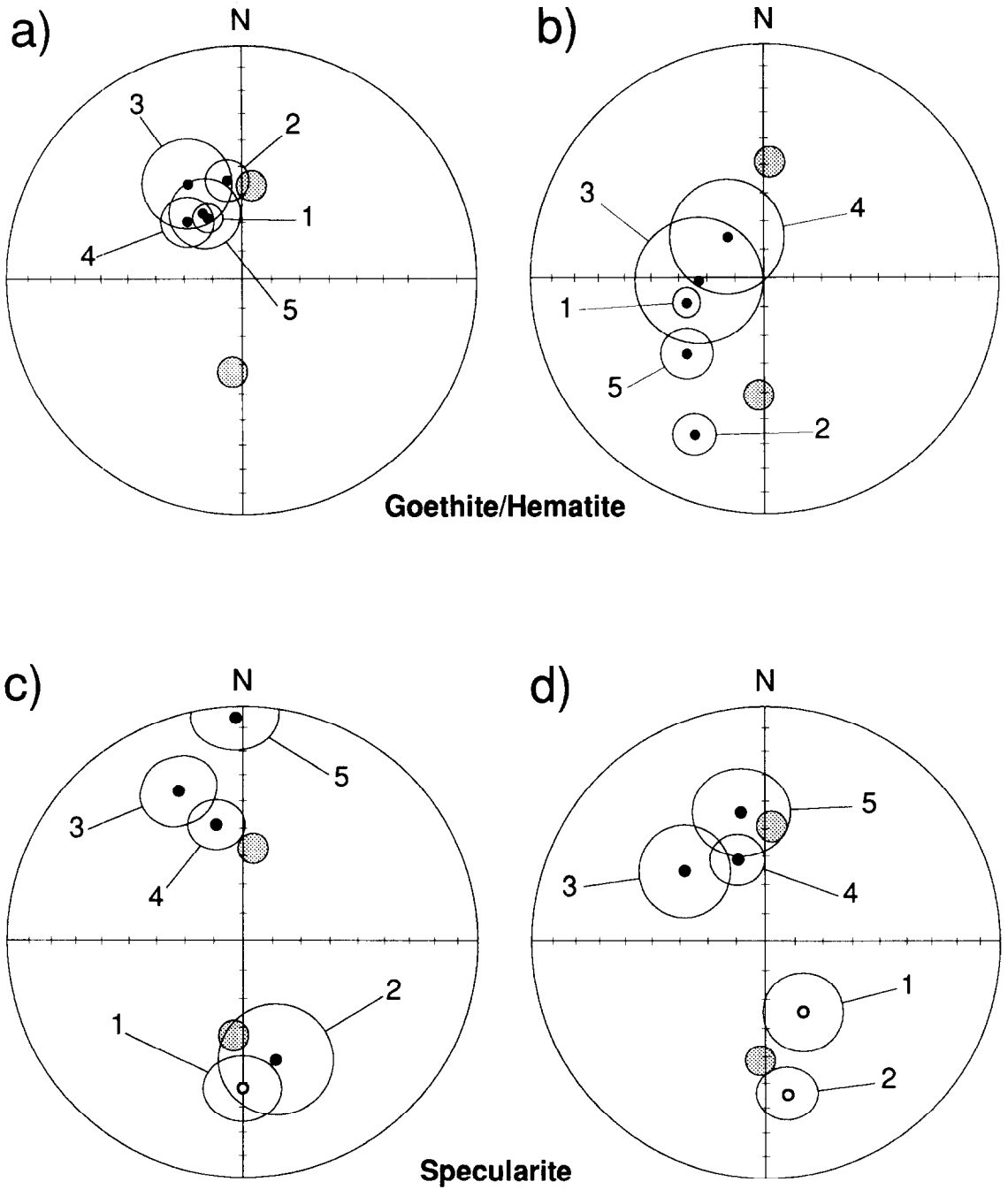


Fig. 5. Equal-area projection of the mean remanence directions of the Berga Formation in the Sierra de Busa (Fig. 2): Goethite/hematite component (a) before tectonic correction, and (b) after tectonic correction. Specularite component (c) before tectonic correction, and (d) after tectonic correction. Dots (open circles) represent directions plotted on lower (upper) hemisphere. The coeval ADF is symbolised by shaded circles, whereas the expected directions for (a, c) the actual ADF, and (b, d) the Early Oligocene (Table 2) correspond to the centre of the circle.

after tectonic correction has a declination of 340.4° and an inclination of 50.5°.

4.2. The Bona Formation

The average intensity of the NRM in the reddish limestone is 1.45 mA m^{-1}. The thermal demagnetisation of the NRM is characterised by a northerly downward-directed component, which is removed below 100°C. A second component, with maximum unblocking temperature of 675°C, trends to the origin of the orthogonal vector plot and shows often negative inclinations (Fig. 6). The IRM of the reddish limestone is dominated by high-coercivity minerals, which were not saturated by the maximum field applied (Fig. 7a). The thermal demagnetisation of the soft (0.12 T) IRM component (Fig. 7b) indicates no significant contribution of low-blocking-temperature minerals to the NRM of the reddish limestone. The thermal demagnetisation of the hard (1 T) IRM component shows two major unblocking steps: one below 100°C and the other between 600 and 625°C (Fig. 7b). These unblocking temperatures are typical of goethite and hematite, respectively (Lowrie and Heller, 1982). Therefore, these two mineral phases are to be considered as the principal remanent magnetisation in these rocks.

The goethite component in the reddish limestone of the Bona Formation shows mean directions which, before tectonic correction, scatter around the present axial dipole field (ADF) (Table 1; locations 14–16). After tectonic correction, dispersion of the directions increases between the locations, and the directions deviate from any expected ADF for Spain since the Late Cretaceous.

The hematite component carries directions which, before tectonic correction, are rotated, and which have shallow inclinations compared with the present ADF direction. Application of the tectonic correction increases the scatter of the declinations, and leads to anomalously shallow inclinations compared to the Late Cretaceous ADF for Spain (Fig. 8b). Also the comparison of the *f*-parameters (McFadden and Jones, 1981) of

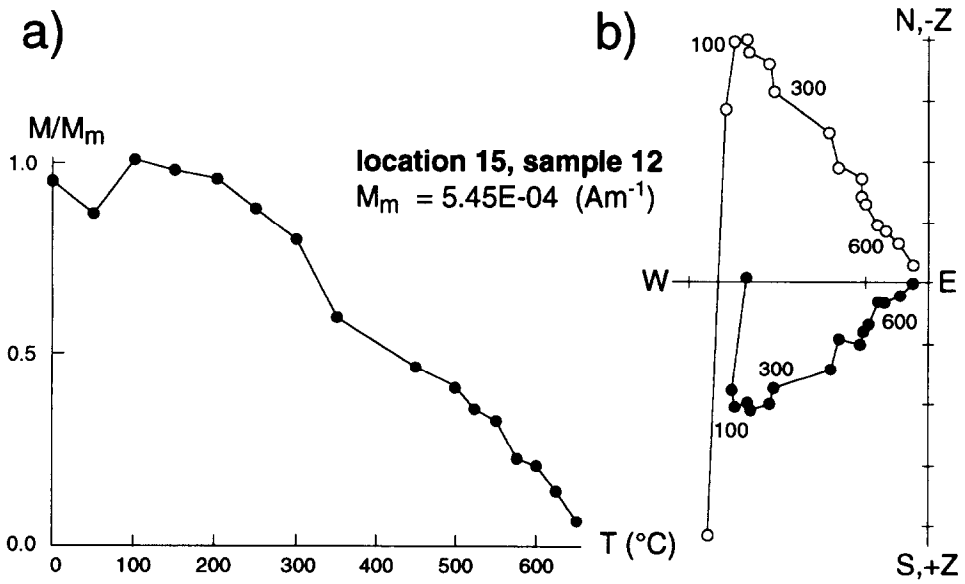


Fig. 6. (a) Decrease in intensity and (b) vector plots of the natural remanence (before tectonic correction) during thermal demagnetisation of a reddish limestone from the Bona Formation. Full and open symbols present projections of the magnetisation vector on the horizontal and E–W vertical plane.

the mean direction before and after tectonic correction (Table 1) provides no unambiguous clues to the origin of magnetisation carried by hematite.

5. Timing of magnetisation and tectonic rotation

In the Ebro Basin, syn-tectonic sediments, outcropping in the Sierra de Busa syncline (Figs. 1b and 2; locations 1–5) and the Puigreig anticline (Fig. 1b; location 6), carry rotated remanence directions of pre- and post-folding origin. The magnetisation directions of goethite/hematite shows a better grouping before than after tectonic correction (Table 1; Fig. 5a and b), and is therefore younger than middle Oligocene (Fig. 2). These magnetisation directions reveal anticlockwise tectonic rotation about vertical axes of about 25° compared to any axial dipole field (ADF) since the Oligocene. The remanence, stable at high temperatures, was acquired before folding of the sand- and siltstone in the Sierra de Busa (Fig. 5c and d), and that points to a syn-sedimentary origin which is likely to be carried by a detrital specularite. Comparison of the directions of this component with the Late Eocene ADF reveals the same anticlockwise rotation of about 25°.

Since both the pre- and the post-folding magnetisation directions have the same deviation from the expected directions, tectonic rotation of the Sierra de Busa and the Puigreig folds can be postulated for times later than middle Oligocene (Table 1; Figs. 1b, 5a and d). For the further evaluations of tectonic rotation only the post-folding directions were considered (Table 2).

The observed anticlockwise rotation of the Sierra de Busa and the Puigreig folds in the Ebro Basin agrees well with data reported from the southern part of the Cadí Unit in the Oliana anticline (Table 2; Fig. 1b; locations 7–9) and east of the Pedraforca Unit (Table 2; Fig. 1b; locations 10–13). In contrast, published data from the Cadí Unit north of the Pedraforca Unit exhibits different magnetisation directions (Table 2; Fig. 1b; locations 24, 25) which can be compared to directions from the Axial Zone (Table 2; Fig. 1b; locations 18–23). This observation indicates divergent tectonic evolution of the northern and the southern parts of the Cadí Unit since the Early Oligocene. Therefore the northern part together with the Axial Zone can be considered as a separate tectonic unit which is referred to as the *Internal Unit* of the East Pyrenees. In analogy, the southern part of the Cadí Unit together with the folded Ebro Basin can be comprehended

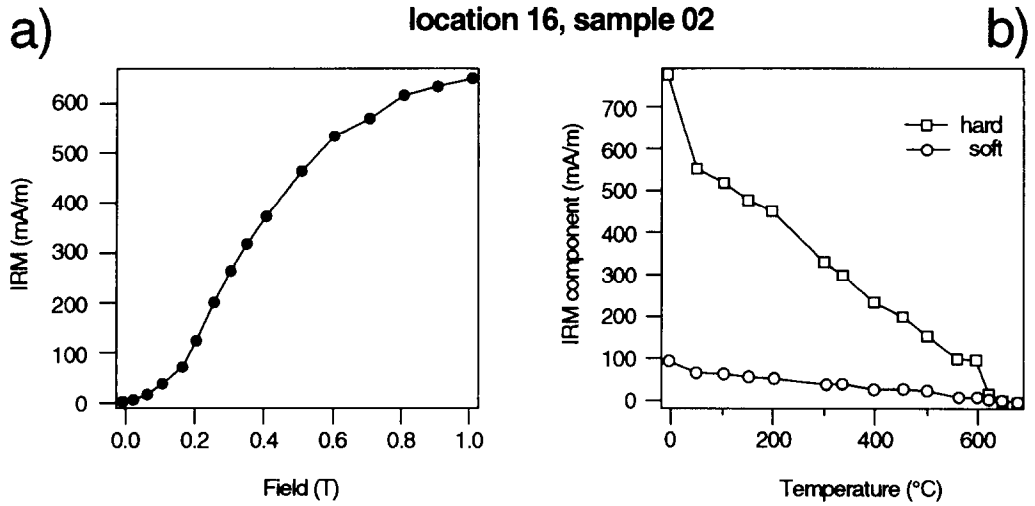


Fig. 7. (a) IRM acquisition and (b) thermal demagnetisation of a two-component IRM of a reddish limestone from the Bona Formation. The two-component IRM is produced by magnetising the sample in an 1 T field (hard) along its z-axis, and in a 0.12 T field (soft) along its x-axis (Lowrie, 1990).

as the *External Unit*. The southern margin of the External Unit is defined by the Samaüja and Suria anticlines (Fig. 1b). This interpretation is supported by paleomagnetic data from the southern Ebro Basin which show different magnetisation directions (Parès, 1989). The border between the Internal and External units can be located, based on the palaeomagnetic data, north of locations 10–13 and south of locations 24 and 25, probably along the Cadí fault (Fig. 1b).

In the Pedraforca Unit both magnetisation directions of the reddish Late Cretaceous limestone show no pre-folding magnetisation (Table 1; locations 14–16). The remanence directions of goethite before tectonic correction are parallel to the present ADF and thus the magnetisation can be explained by means of sub-recent chemical weathering. The directions carried by hematite, before tectonic correction, show eastwards rotated declinations (Table 1; Fig. 8), and inclinations too shallow compared to the present ADF. Similar magnetic properties and directions have been found in reddish limestone of the same formation from the Sierra d'Ensija (Fig. 1b; location 18). Keller and Gehring (1992) demonstrated that the hematite magnetisation in this reddish limestone is a product of Late Eocene weathering by correcting bedding of conglomerates to reconstruct the palaeoweathering horizon. At locations 14–16 (Fig. 1b), the bedding of the conglomerates overlaying deformed reddish limestone is horizontal, and therefore no tilt correction is necessary. The inclinations of these locations agree well with the Late Eocene ADF (Fig. 8a). All hematite magnetisation directions in this weathered limestone, however, show a clockwise rotation of about 30° from the direction expected for the Late Eocene (Fig. 8a). The distribution of the locations allows to interpret the deviation of the remanence directions as large-scale tectonic rotation that probably included the whole Pedraforca Unit. Since the magnetisation is associated with Late Eocene weathering, the rotation of the Pedraforca Unit has to be younger. By contrast, palaeomagnetic data from the tectonically equivalent South Central Unit, west of the STZ (Fig. 1), exhibit no rotation since the Eocene (Dinarès et al., 1992; Pascual et al., 1992).

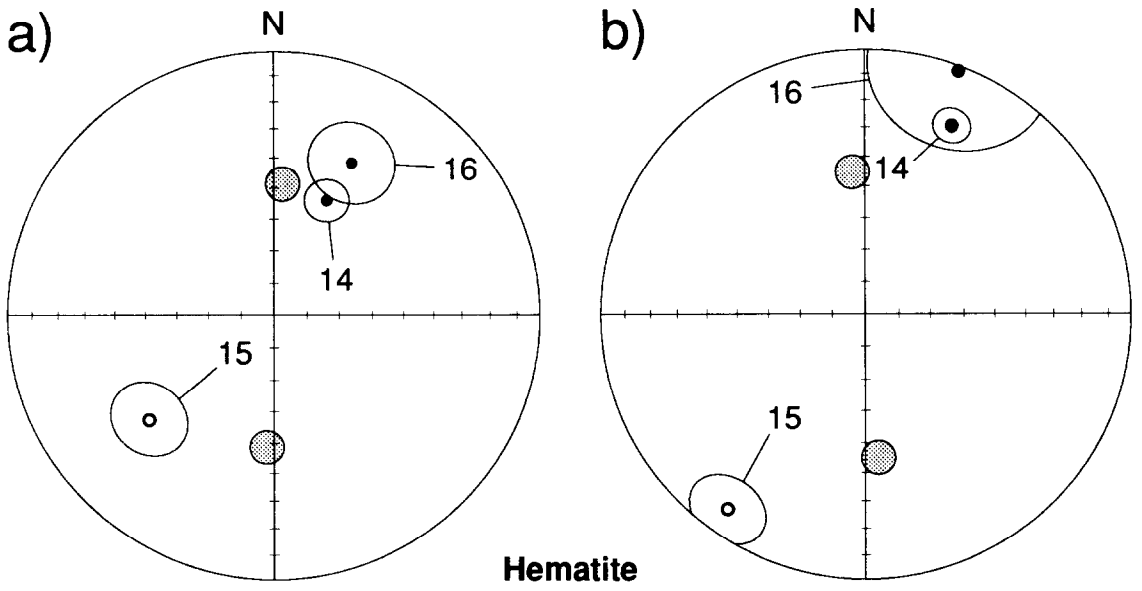


Fig. 8. Equal-area projection of the mean hematite magnetisation directions of locations 14–17 from the reddish limestone of the Bona Formation: (a) before tectonic correction, (b) after tectonic correction. The coeval ADF is symbolised by shaded circles, whereas the expected directions for (a) the Late Eocene (Table 2) and (b) the Late Cretaceous (after Van der Voo and Zijderveld, 1971) correspond to the centre of the circle.

The magnetic data from the Southeast Pyrenees were obtained from different stratigraphic levels from Permian to Oligocene (Table 2). In this time interval considerable apparent polar wander took place. To evaluate the data, each mean direction was compared to the coeval expected direction calculated for the locations from appropriate palaeopoles. The differences (ΔD) of the observed declinations and the differences (ΔI) of the observed inclinations from the declinations and inclinations, respectively, of the expected directions were computed and averaged for each tectonic unit (Table 2). The insignificant mean ΔI values indicate that the performed calculations are adequate.

In the Internal Unit the observed declinations scatter around the expected declinations which results in a mean ΔD close to zero ($\Delta D_I = -1°$). Within the Pedraforca and External units, the declinations show systematic deviations from the expected declinations. The deviation is clockwise in the Pedraforca Unit ($\Delta D_P = 32°$) but anti-

Table 2
Calculation of deviations between observed and expected directions

Location	Reference	Mag. age	Expected Dec/Inc	Observed Dec/Inc	α_{95}	$\Delta D/\Delta I$	Unit mean
External Unit							
1	this study	eO	006/57	331/65	5.0	−35/8	
2	this study	eO	006/57	352/54	7.6	−14/−3	
3	this study	eO	006/57	330/52	15.8	−36/−5	
4	this study	eO	006/57	329/64	12.4	−37/7	
5	this study	eO	006/57	317/62	8.8	−49/5	
6	this study	eO	006/57	329/53	15.7	−37/−4	ΔD_E: −25, σ_E: 13
7	DN92	mE	003/49	350/32	7.2	−13/−17	ΔI_E: −6, σ_E: 10
8	DN92	mE	003/49	342/28	6.8	−21/−21	
9	DN92	mE	003/49	337/27	15.5	−26/−22	
10	BU92	mE	003/49	354/40	4.1	−9/−9	
11	BU92	mE	003/49	346/48	4.4	−17/−1	
12	BU92	eE	003/49	344/48	5.5	−19/−1	
13	BU92	mE	003/49	355/40	4.7	−8/−9	
Pedraforca Unit							
14	this study	lE	003/49	025/51	6.9	22/2	
15	this study	lE	003/49	051/40	12.4	48/−9	ΔD_P: 32, σ_P: 11
16	this study	lE	003/49	027/35	6.8	24/−14	ΔI_P: −8, σ_P: 7
17	KG92	lE	003/49	038/40	5.5	35/−9	
Internal Unit							
18	VD67	eP	331/−4	350/03	6.0	19/7	
19	VD67	lP	345/12	342/04	8.0	−3/−8	
20	VD67	lP	345/12	333/33	10.0	−12/21	
21	VD67	eT	337/22	341/24	11.0	4/2	ΔD_I: −1, σ_I: 11
22	KE92	P/T	337/22	327/28	15.4	−10/6	ΔI_I: 4, σ_I: 10
23	KE92	P/T	337/22	325/23	20.0	−12/10	
24	DN92	mE	003/49	012/39	11.5	9/−1	
25	KE92	lE	003/49	003/61	13.2	0/12	

Data are from this study (rounded to integers) and from Dinarès et al. (1992) (DN92), Burbank et al. (1992) (BU92), Keller and Gehring (1992) (KG92), Keller (1992) (KE92) and Van Dongen (1967) (VD67). All NRM directions are converted into lower hemisphere polarity. The expected directions were calculated for the sampling location using published palaeopoles for the age of magnetisation (Mag. age): Early Permian (eP) and Late Permian (lP) palaeopoles are from Van der Voo and French (1974), Permotriassic (P/T) and Early Triassic (eT) from VandenBerg and Zijderveld (1982), Late/Early Eocene (l/eE) and Early Oligocene (eO) from Westphal et al. (1986). ΔD and ΔI are differences in declination and inclination, respectively, between the observed and the expected directions, σ_D and σ_I are the corresponding standard deviations.

clockwise in the External Unit ($\Delta D_E = -25°$). The mean ΔD of the three units reveal significant discrepancies in declination which can be explained by contrasting tectonic rotation about vertical axes of the Pedraforca and External units (Table 2).

6. Implications for the structural interpretation of the Southeast Pyrenees

The tectonic rotation in the External Unit (Table 2) is younger than folding of the Sierra de Busa syncline (Fig. 2) and the Puigreig anticline (Early Oligocene; Riba, 1976; Sáez and Riba, 1986). This anticlockwise rotation can be explained by differential compression during the latest stage of Pyrenean thrusting. From balanced cross sections, Vergés and Burbank (1994) calculated for the middle to Late Oligocene a decrease in shortening from about 17 km at Oliana to zero east of Berga. The differential compression caused several NE–SW-striking anticlines in the Ebro Basin (e.g., Cardona and Suria anticlines; Fig. 1b), and led to rotation of the Puigreig–Oliana anticline (Dinarès et al., 1992; Keller, 1992). The NW–SE-striking Samaüja anticline, the southwestern margin of the External Unit, is in this scenario explained as the result of right-lateral transpression due to anticlockwise rotation of the foreland (Fig. 1b).

The Pedraforca Unit was thrust onto the sediments of the External Unit during the Early Eocene (Martínez et al., 1988). The magnetisation in the palaeoweathering horizon of the Pedraforca Unit was dated as Late Eocene (Keller and Gehring, 1992). Consequently, the observed tectonic rotation (Table 2) is younger than the emplacement of the Pedraforca Unit. Since the Pedraforca lies on the External Unit, and the two units show different senses of rotation, it follows that the overall rotation of the Pedraforca Unit may well have been 57° (Table 2). To explain this rotation post-Late Eocene tectonics have to be taken into account.

During the Late Eocene to Early Oligocene compressional tectonics in the Southeast Pyrenees resulted in thrusting of the northern and southern part of the Cadí Unit. Subsequent shortening (middle to Late Oligocene) led to anticlockwise rotation of the southern part together with the adjacent Ebro Basin, and to the formation of the External Unit. During this compressional phase the Pedraforca Unit was piggy-back transported on the External Unit (cf. Muñoz et al., 1986; Vergés et al., 1992). Since the External Unit shows an anticlockwise rotation and no rotation was found in the Internal Unit, differential shortening as a possibility to explain the clockwise rotation of the Pedraforca Unit can be excluded. Following this argument the clockwise rotation can not be explained by Pyrenean thrust tectonics and is therefore younger than Late Oligocene.

Pull-apart basins of Neogene age (e.g., Cerdanya Basin; Fig. 1) have been reported from the Axial Zone north of the Southeast Pyrenees (Pous et al., 1986). The formation of these basins has been explained by right-lateral tectonic movements (Cabrera et al., 1988). In the Southeast Pyrenees the Cadí fault, the northern border of the Pedraforca Unit, has been interpreted as Neogene fault with a vertical offset in a kilometre range (Fig. 1b; Ullastre et al., 1987; Durand-Delga et al., 1989). The southern border formed by the Vallfogona fault (Fig. 1a) is 50–70° steep (cf. Solé-Sugrañes, 1973; Muñoz et al., 1986), what indicates a fault parallel slip component. Thus, this fault can be considered as the southern margin of the right-lateral movements in the East Pyrenees. Clockwise rotation, however, is often observed related to right-lateral tectonics (e.g., Freund, 1974; Ron et al., 1984). Therefore it can be postulated that the clockwise rotation of the Pedraforca Unit, situated between the Cadí and the Vallfogona faults, is the result of Neogene right-lateral wrench tectonics.

7. Conclusions

Palaeomagnetic data from the Southeast Pyrenees lead to the following conclusions:

(1) According to their palaeomagnetic characteristics the Southeast Pyrenees can by subdi-

vided into the Internal, External and Pedraforca units.

(2) The Pedraforca Unit shows clockwise tectonic rotation of 57° compared to the External Unit which itself shows an anticlockwise rotation of 25° compared to the Internal Unit.

(3) The anticlockwise rotation of the External Unit is explained by differential shortening during middle to Late Oligocene.

(4) The clockwise rotation of the Pedraforca Unit is indicative of Neogene right-lateral wrench tectonics, which suggests a different tectonic regime compared to Pyrenean thrusting.

(5) The timing and sense of post-thrusting tectonic rotation found in the Southeast Pyrenees agrees well with the opening of pull-apart basins in the Axial Zone. This suggests that the post-thrusting tectonics could have also modified the thrust geometry elsewhere in the Pyrenees.

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